

Readout-compatible CA1 plasticity supports cue-dependent reconfiguration and degraded-cue recall in a minimal hippocampal model

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Abstract

Rapid episodic memory requires not only selecting stored content, but expressing that content in a form that downstream circuits can interpret. We examine this constraint in a minimal hippocampal model in which entorhinal cortex (EC) generates a CA3-like retrieval key and an instructed CA1 state, plastic CA3–CA1 weights associate the key with that state, and a fixed pretrained CA1–EC decoder reconstructs the input. Pretraining establishes a representational basis before rapid associative learning. Across 20 paired initializations, a fixed permutation of the instructed CA1 coordinates impaired one-shot reconstruction under both an instructive-driven rule (mean output–target cosine, 0.804 aligned versus 0.125 permuted) and a bounded error-driven rule (0.514 versus 0.066). Applying the corresponding permutation to the decoder restored performance, demonstrating that preserved information can become unreadable to an unchanged output pathway. On a cue-guided circular track, exchanging cue identities caused abrupt, reproducible changes in CA1 tuning relative to a matched no-swap control. Both rules generated continuous mixtures of spatially stable and cue-modulated responses, while differing quantitatively in one-shot reconstruction, tuning stability, and position decodability. During frozen recall, selective degradation of lateral or medial EC inputs preferentially impaired cue or position readout, respectively, while the complementary cue readout was preserved under medial EC degradation. A dense common-key control failed to distinguish cue or position states. These simulations support a limited proposal: CA3-like keys can select associative content, while rapid CA1 plasticity expresses that content within a structured basis that remains legible to an established readout. The model demonstrates degraded-cue robustness, not recurrent CA3 pattern completion or long-term stability.

1 Introduction

The hippocampus is commonly described as a fast-learning system that binds elements of an experience and later retrieves them from partial input. Hippocampal indexing and complementary-learning-systems theories formalize this role as rapid episodic storage operating alongside slower cortical learning [14, 19]. An effective memory system must therefore separate related episodes, select an appropriate stored state, and make the retrieved content usable to downstream circuits. The first two requirements are central to classical theories of pattern separation and completion. Here we focus on the third: the representational coordinates in which retrieved content is expressed. This motivates a view of CA1 not merely as a relay or container of individual memories, but as an adaptive interface that organizes successive experiences within a structured neural space.

The anatomy of the hippocampal formation suggests a distribution of computational roles across its subfields. Entorhinal cortex (EC) provides major input to the hippocampus. Medial EC (MEC) is

strongly associated with spatial variables, including grid-cell and position-related activity, whereas lateral EC (LEC) more prominently represents objects, sensory features, and salient events, although the division is neither absolute nor anatomically strict [3, 7, 8]. The canonical trisynaptic pathway projects through dentate gyrus to recurrent CA3 circuitry and then, through Schaffer collaterals, to CA1; CA1 also receives direct EC input through the temporoammonic pathway. Autoassociative theories assign CA3 a central role in content-addressable retrieval [13]. Consistent with this view, selective disruption of CA3 NMDA-receptor function impairs recall from partial cues, and CA3 and CA1 ensembles exhibit distinct responses to environmental change [9, 17]. More recent models develop recurrent CA3 dynamics and spatially organized associative scaffolds [2, 11]. These accounts explain how a memory may be selected, but do not by themselves determine how the selected content should be expressed at hippocampal output.

CA1 representations can form and reorganize rapidly. Behavioral-timescale synaptic plasticity (BTSP) can create a place field after a single dendritic plateau event and can bidirectionally reshape an existing field according to its relation to the instructive event [1, 15]. EC layer 3 activity can provide an instructive signal for learning-related changes in CA1 population activity [6]. Recent experiments and models further connect BTSP-like dynamics to trial-by-trial field shifts and the progressive stabilization of task-relevant spatial codes [12, 20]. BTSP is therefore relevant not only to inserting an isolated place field, but also to updating a cognitive map as cues and task contingencies change. Computational work additionally shows that BTSP-like rules can support rapid content-addressable storage [22]. The present study abstracts this broad motif of an EC-related instruction acting on CA3–CA1 synapses; it does not model the temporal eligibility trace, plateau potential, or molecular mechanisms of biological BTSP.

CA1 activity is also context sensitive. Changes in environmental or cue configuration can alter firing rates, field positions, or the active population, producing phenomena commonly described as rate and global remapping [10, 16]. Direct entorhinal input offers a route by which spatial and nonspatial variables can jointly shape these changes. In the model below, separate MEC-like and LEC-like input partitions provide a controlled way to manipulate those variables. This is a modeling abstraction used to test pathway-selective perturbations, not a claim that biological MEC and LEC carry pure or independent information channels.

Our central proposal adds a representational constraint to this circuit motif. If rapid plasticity writes an arbitrary CA1 pattern while its output pathway remains fixed, the pattern may retain information about an episode yet be functionally unreadable because its coordinates no longer have the meaning expected by the recipient circuit. Related constraints have been demonstrated outside the hippocampus. In brain–computer-interface experiments, motor cortical perturbations that remained within a pre-existing neural manifold were learned more readily than perturbations requiring activity outside that manifold [18]; short-term adaptation could also occur by reassociating an established activity repertoire with new outputs [5]. Stable latent population dynamics can support behavioral decoding despite turnover of individual recorded neurons [4]. We use the narrower term *readout compatibility* for the requirement that a newly instructed CA1 state remain expressed in coordinates that an established output pathway can interpret.

To instantiate prior structure, an EC–CA1–EC autoencoder is pretrained before new associations are stored. This is a computational device, not a claim that CA1 is biologically optimized by backpropagation in a separate phase. Biologically, it abstracts structure supplied by development, accumulated experience, and relatively stable entorhinal–hippocampal connectivity [2, 14, 21]. In deep learning, pretraining plays an analogous engineering role by establishing reusable features

before rapid adaptation, although the mechanisms and substrates differ [23]. Freezing the decoder after pretraining turns compatibility into an explicit assay: a new CA1 state must remain legible to the established readout rather than being rescued by encoder–decoder co-adaptation. Such a relation could, in principle, preserve the decodability of successive experiences over longer timescales, but duration itself is not tested here.

We study this idea in a deliberately minimal model. EC input generates both a discriminable CA3-like retrieval key and a pretrained instructed CA1 state. Plastic CA3–CA1 synapses associate the key with that state, and a frozen CA1–EC decoder reads out recall. We compare an instructive-driven rule with a bounded error-driven rule that corrects the discrepancy between instructed and recalled CA1 activity. The simulations ask whether reconstruction depends on coordinate compatibility, whether cue-sequence changes induce reproducible CA1 reconfiguration, how the two update rules differ under matched conditions, and how recall changes when spatial or cue-like EC input is selectively degraded. The resulting claims are constructive and conditional: the model proposes a representational principle and exposes its consequences, but does not establish that biological CA3 or CA1 implements the exact architecture, pretraining procedure, or synaptic rules used here.

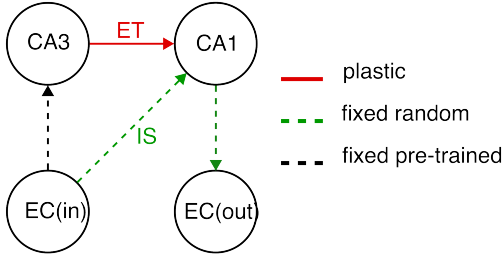
2 Results

2.1 A minimal retrieval-key and instructed-CA1 architecture

The model contains four functional components (Fig. 1A). Entorhinal input, denoted the EC target (ET), projects through a fixed sparse map to a CA3-like population that provides a retrieval key. A separately pretrained EC–CA1 encoder transforms the same input into an instructed CA1 state (IS). Plastic CA3–CA1 weights associate the retrieval key with the IS, whereas the corresponding pretrained CA1–EC decoder remains fixed. During recall, the EC input regenerates a key, the learned association reconstructs a CA1 state, and the decoder maps that state back to EC output. This architecture separates the selection of associative content from its expression in a pre-existing output basis.

We evaluated two plasticity rules under matched inputs and parameters (Fig. 1B). The instructive-driven rule writes an outer-product association gated by the IS without considering the current recalled state. The bounded error-driven rule instead potentiates CA3–CA1 synapses for underexpressed CA1 components and depresses them for overexpressed components. Repeated presentations therefore rewrite the instructed association under the instructive-driven rule but correct a residual under the error-driven rule. Both are abstract models of rapid instructed plasticity; neither implements the temporal eligibility trace or dendritic plateau dynamics of biological BTSP.

Track inputs combined a MEC-like spatial component with a LEC-like cue component (Fig. 1D). Two cue identities occurred at fixed positions and could exchange locations across blocks. For degradation tests, the model first stored intact laps; plasticity was then frozen while a fixed fraction of either the MEC-like or LEC-like input units was set to zero (Fig. 1E).

A**B**

$$\text{instructive-driven} \quad W(t+1) = \alpha x_{\text{IS}} x_{\text{ET}}^T + (1 - \alpha x_{\text{IS}}) \odot W(t)$$

$$\begin{aligned} \text{error-driven} \quad & e^+ = [x_{\text{IS}} - x_{\text{CA1}}]_+ \\ & e^- = [x_{\text{CA1}} - x_{\text{IS}}]_+ \\ & \delta^+ = \alpha (e^+ \cdot x_{\text{CA3}}^T) (1 - W(t)) \\ & \delta^- = \alpha (e^- \cdot x_{\text{CA3}}^T) W(t) \\ & W(t+1) = \delta^+ - \delta^- + W(t) \end{aligned}$$

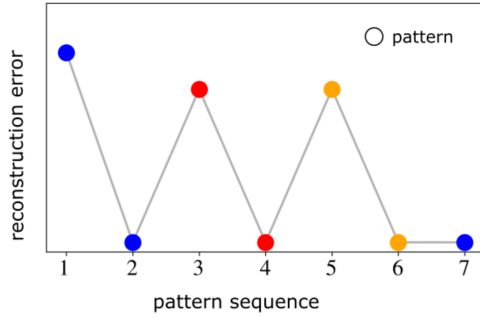
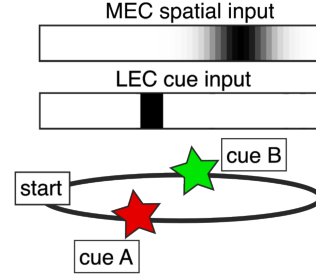
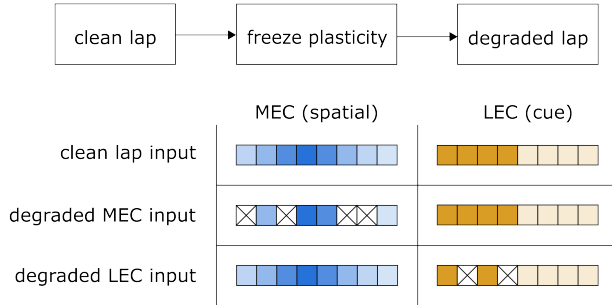
C**D****E**

Figure 1: Model architecture and simulation protocol. (A) EC input (ET) produces a fixed sparse CA3-like retrieval key and, through a pretrained encoder, an instructed CA1 state (IS). Plastic CA3–CA1 weights store their association; a fixed pretrained CA1–EC decoder reads out recall. (B) Instructive-driven and bounded error-driven update rules. (C) Conceptual reconstruction-error trajectories over first and repeated presentations; colors denote pattern identity. This panel is a schematic rather than a measured result. (D) Circular-track input composed of MEC-like spatial activity and LEC-like cue activity. (E) Selective degradation protocol. Intact laps are stored before plasticity is frozen; a fraction of MEC-like or LEC-like units is then zeroed using a mask held fixed within a probe lap and resampled across probes.

2.2 Readout compatibility constrains rapid associative storage

We first isolated the relationship between an instructed CA1 code and its fixed output decoder (Fig. 2A). Twenty independently initialized autoencoders were pretrained on sparse EC patterns and then frozen. For each initialization, 28 held-out patterns were stored once under five paired conditions. In the aligned condition, the IS remained in the coordinates learned by the decoder. In the fixed-permutation condition, the same non-identity permutation was applied to every IS while the decoder was left unchanged. In the matched-decoder condition, the corresponding decoder columns were permuted together with the IS. Random-content and no-plasticity conditions controlled for storing an incorrect association and for leaving the memory unwritten, respectively.

Under the instructive-driven rule, aligned instructions produced a mean output–target cosine of 0.804 (95% confidence interval [CI], 0.783–0.825), whereas a fixed coordinate permutation reduced it to 0.125 (0.114–0.135). The paired aligned-minus-permuted difference was 0.679 (0.655–0.704). The same coordinate dependence appeared under the error-driven rule: aligned and permuted means were 0.514 (0.493–0.535) and 0.066 (0.050–0.082), respectively, giving a paired difference of 0.448 (0.418–0.477). Permuting the decoder together with the IS restored the aligned mean exactly for each rule. Random-content recall remained low (0.189 for the instructive-driven rule and 0.092 for the error-driven rule), as did the shared no-plasticity control (0.191). The matched-decoder restoration is an algebraic coordinate control: the permutation preserves the IS values but changes their meaning to the unchanged decoder. Reconstruction therefore depends on compatibility between the instructed CA1 coordinates and the fixed output basis, not on information content alone. The higher one-shot aligned recall under the instructive-driven rule is also consistent with its full instructed write, whereas the error-driven rule makes a bounded residual correction at the same learning rate.

2.3 Cue exchanges induce reproducible CA1 reconfiguration

We next exposed the model to 40 track laps in which two cue identities exchanged positions every 10 laps (Fig. 2B). A paired no-swap control used the same root seeds, pretrained encoder and decoder, CA3 wiring, MEC-like trajectories, lap count, and plasticity parameters, but retained the initial cue assignment throughout. After every training lap, plasticity was paused and the model was probed in the context scheduled for that lap.

For the instructive-driven rule, mean unit-wise tuning similarity between consecutive probes fell to 0.365 at cue exchanges (95% CI, 0.348–0.382), compared with 0.850 at the same lap boundaries in the no-swap control (0.837–0.863). The paired swap-minus-control difference was -0.484 (-0.504 to -0.464). Under the error-driven rule, exchange-boundary similarity was 0.451 (0.436–0.466), compared with 0.824 in the matched control (0.813–0.835), for a paired difference of -0.373 (-0.392 to -0.353). At non-exchange transitions in the alternating schedule, similarity remained high for both the instructive-driven rule (0.833, 0.823–0.843) and the error-driven rule (0.821, 0.814–0.828). The abrupt tuning changes were therefore associated with the altered cue sequence rather than with continued learning alone.

After learning, plasticity was frozen and both cue arrangements were probed. The representative error-driven population contained spatially stable fields, amplitude-modulated responses, and fields whose preferred position changed with cue identity (Fig. 2C). Across seeds, mean unit-wise, mean-centered spatial stability was 0.330 (95% CI, 0.297–0.363) under the instructive-driven rule and 0.464 (0.441–0.486) under the error-driven rule. Mean absolute cue modulation was 0.132 (0.130–0.135)

and 0.124 (0.120–0.128), respectively. In both rules, the unit-level distributions were continuous rather than separated into discrete response classes (Fig. 2D), and strong cue modulation was concentrated among units with low or negative cross-context stability. These distributions are descriptive: units nested within a seed were not treated as independent replicates, and the analysis is not a formal classification of rate versus global remapping. Operationally, changing the cue arrangement changed both the EC-derived instruction and the CA3-like key, producing a mixture of stable and context-dependent recalled CA1 activity.

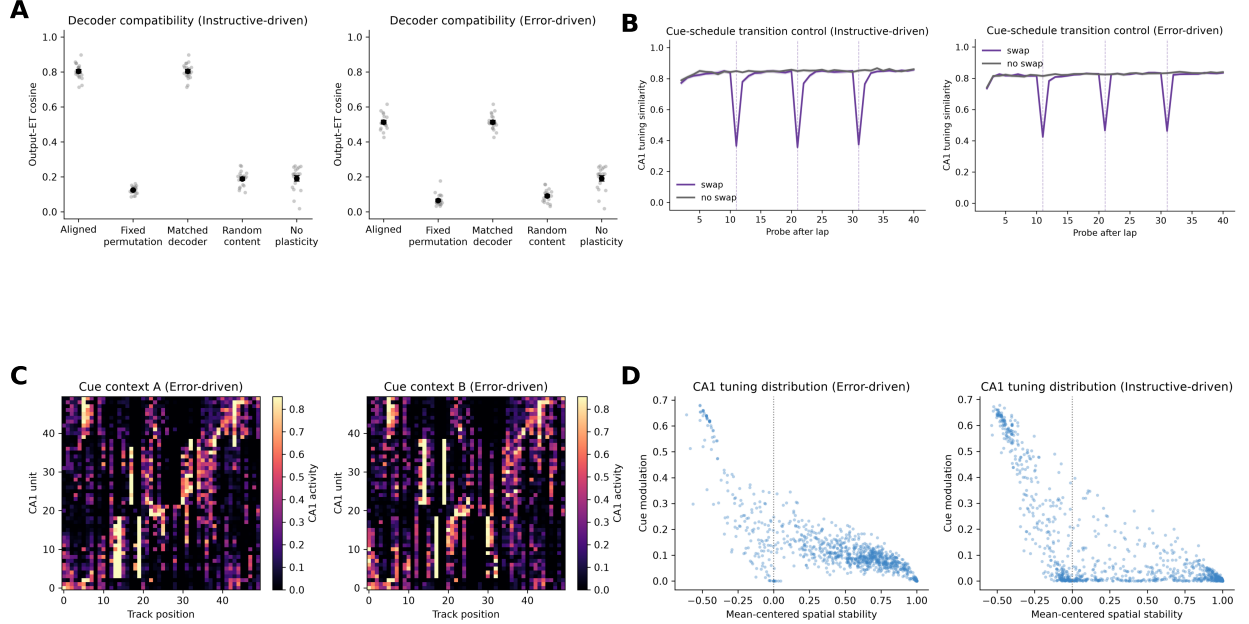


Figure 2: Readout compatibility and cue-dependent CA1 reconfiguration. (A) Output–target cosine in aligned, fixed-permutation, matched-decoder, random-content, and no-plasticity conditions for the instructive-driven rule (left) and error-driven rule (right). Grey points are seed means across 28 stored memories; black points and error bars show mean $\pm$ SEM across 20 seeds. (B) Mean unit-wise tuning similarity between consecutive scheduled-context probes for the instructive-driven rule (left) and error-driven rule (right). Purple denotes the cue-exchange schedule and grey the matched no-swap control; vertical dashed lines mark exchanges. Curves show mean $\pm$ SEM across 20 paired seeds. (C) Recalled CA1 activity on the same track under cue contexts A (left) and B (right) for the representative error-driven seed. Units have the same order and are sorted by preferred position across contexts; color limits are shared. (D) Unit-level relationship between mean-centered spatial stability and cue modulation for the error-driven rule (left) and instructive-driven rule (right), pooling 50 units from each of 20 seeds (1,000 points per rule). The vertical line marks zero stability. Panels C and D illustrate heterogeneous stable and cue-dependent responses but do not formally classify remapping type; confidence intervals in the text use seed-level summaries.

2.4 Selective EC degradation dissociates cue and spatial readout

We then asked whether the stored association remained usable when one EC input modality was selectively corrupted (Fig. 3). Each model stored 20 intact laps with a fixed cue arrangement. Plasticity was frozen, and independent masks removed 0, 25, 50, 75, or 90% of eligible units. Each fraction used 12 masks per seed and 20 paired root seeds. The principal control replaced the normal sparse EC–CA3 map with a dense common map in which every CA3-like unit received the same average EC input. This manipulation removes item-specific key structure and is therefore a

degenerate-key, not a sparsity-matched, control.

Under LEC-like degradation, cue identity accuracy for the normal key declined from 1.000 to 0.871 at 90% dropout under the instructive-driven rule (95% CI at the endpoint, 0.803–0.938) and from 1.000 to 0.829 under the error-driven rule (0.766–0.893; Fig. 3A). The corresponding endpoint changes were -0.129 (-0.197 to -0.062) and -0.171 (-0.234 to -0.107). MEC output–target cosine remained near its instructive-driven baseline (0.736 at baseline and 0.723 at 90% dropout; endpoint change -0.013 , -0.029 to 0.003), whereas it declined from 0.904 to 0.780 under the error-driven rule (change -0.125 , -0.136 to -0.113 ; Fig. 3B). Thus spatial output remained substantially readable under LEC-like corruption, although it was not strictly invariant under the error-driven rule.

MEC-like degradation produced the complementary loss of position information. Nearest-position accuracy declined from 0.163 to 0.070 under the instructive-driven rule (endpoint change -0.093 , 95% CI -0.106 to -0.079) and from 0.666 to 0.069 under the error-driven rule (change -0.597 , -0.636 to -0.557 ; chance = 0.02; Fig. 3C). Cue identity remained 1.000 for every seed and degradation level under both rules (Fig. 3D). The low baseline position accuracy under the instructive-driven rule is itself informative: repeated error-driven correction yielded a substantially more discriminable spatial readout before degradation, whereas the instructive-driven rule did not. The two rules should therefore not be interpreted as starting from equivalent clean spatial performance.

The dense common-key control remained at chance for cue identity (0.5) and position (0.02) even without degradation, and produced low spatial-output cosine. Its failure establishes that a non-specific common key cannot support the associations, but does not isolate sparsity from key discriminability. Together, the normal-key results demonstrate degraded-cue robustness and pathway-selective failure within this architecture. They do not establish recurrent pattern completion: the CA3-like layer has no attractor dynamics, and recall deteriorates once the feedforward retrieval key loses discriminability. Supplementary Fig. S1 collects four seed-paired contrasts that summarize how the rule choice changes these effects.

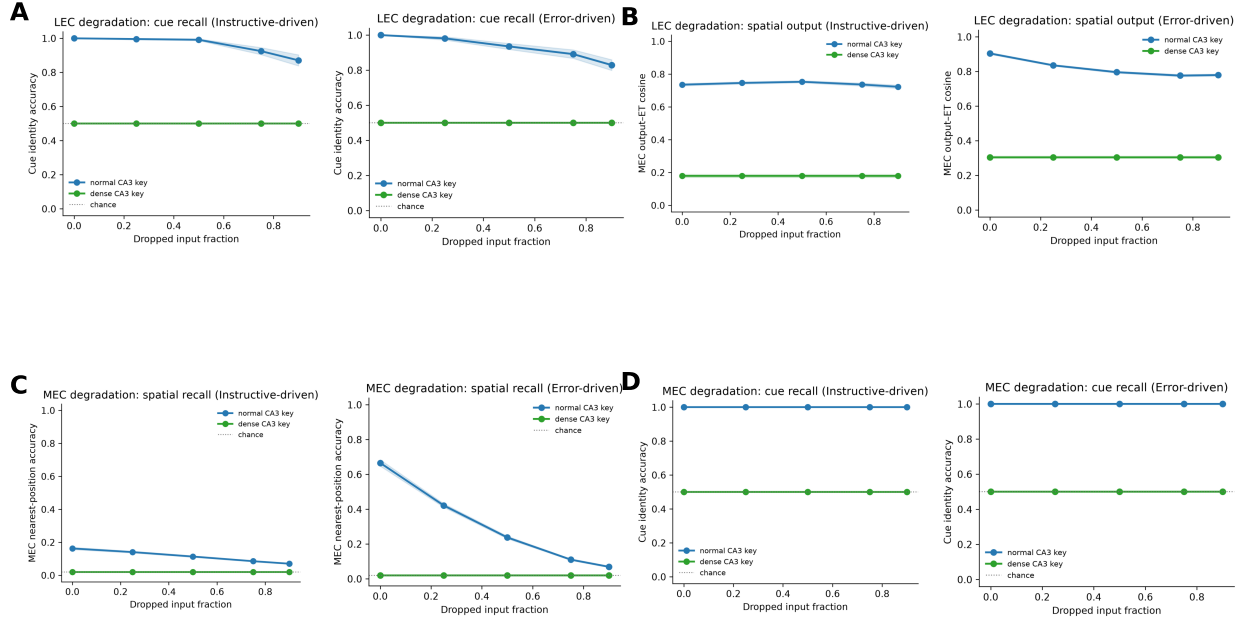


Figure 3: Modality-specific recall under selective EC degradation. Each panel shows the instructive-driven rule on the left and error-driven rule on the right. **(A)** Cue-identity accuracy during LEC-only degradation. **(B)** MEC output–target cosine during LEC-only degradation. **(C)** MEC nearest-position accuracy during MEC-only degradation. **(D)** Cue-identity accuracy during MEC-only degradation. Curves show mean $\pm$ SEM across 20 seeds after averaging 12 masks within each seed. Blue denotes the normal sparse CA3-like retrieval map; green denotes the dense common-key control, which collapses item-specific key structure and is not sparsity matched. Dotted horizontal lines denote chance accuracy where applicable. All degraded probes were performed with plasticity disabled.

3 Discussion

The central result is a representational constraint rather than a new claim about hippocampal capacity. In this minimal architecture, a CA3-like retrieval key selects associative content, plastic CA3–CA1 weights express that content as a CA1 state, and a fixed decoder determines whether the state remains usable at the output. A coordinate permutation preserves the instructed vector’s values yet disrupts reconstruction when the decoder is not transformed with it. The matched-decoder rescue makes the logic exact: rapid plasticity must preserve, or co-adapt with, the representational coordinates assumed by recipient circuits. This motivates viewing CA1 not only as a locus of experience-dependent activity, but also as an adaptive interface that places retrieved content within a structured neural space. Such a space could allow successive experiences to remain decodable even while individual fields and participating neurons change.

This proposal complements rather than replaces existing accounts of hippocampal memory. Autoassociative CA3 theories explain how a stored state can be selected from a partial cue, whereas BTSP models explain how temporally broad instructive events can rapidly alter CA1 synapses [1, 17, 22]. The present model abstracts selection as a non-recurrent CA3-like key and asks what must additionally hold when the selected state is expressed at hippocampal output. The relationship to neural-manifold work is conceptual: those studies concern motor cortical learning, not hippocampal memory, but similarly show that an unchanged readout assigns meaning only to particular population coordinates [5, 18]. Readout compatibility is therefore a general systems constraint instantiated here in a hippocampal circuit motif. The simulations do not test retention across days or years; long-term decodability is a hypothesis about what preserving this relation might enable, not an observed outcome.

Pretraining and rapid plasticity have distinct computational roles in this proposal. Pretraining establishes a sparse representational basis together with the decoder that gives its coordinates meaning; rapid CA3–CA1 learning then places a new, key-addressable experience within that basis. This resembles feature reuse after pretraining in deep networks [23], but the analogy is functional rather than mechanistic. Likewise, the covariance matrix adaptation evolution strategy (CMA-ES) used during preliminary model development was an engineering procedure for locating a workable parameter regime, not a proposed neural learning mechanism. In a biological circuit, the basis could reflect development, prior learning, and ongoing entorhinal–hippocampal organization rather than a separate optimization phase. The frozen decoder is an assay of compatibility, not a claim that biological recipient pathways are immutable. The broader prediction is that changes in a CA1 code and its downstream readouts must remain coordinated.

The cue-track simulations connect readout compatibility to context-dependent reconfiguration. Exchanging cue identities produced abrupt changes in recalled CA1 tuning under both plasticity rules, whereas matched no-swap controls remained stable at the same lap boundaries. After learning, both rules produced continuous mixtures of spatially stable and cue-modulated units. This behavior is qualitatively consistent with cue-sensitive field and rate changes and with evidence that EC-directed BTSP can form, shift, and stabilize CA1 fields around behaviorally relevant features [6, 10, 12, 20]. On this interpretation, instructed plasticity acts as an update operation on a spatial cognitive map: it can insert, weaken, or reposition fields as relevant cues change. The simulations are not a quantitative model of biological remapping, however, and the unit distributions were not compared with recorded field statistics. Their narrower implication is that reconfiguration should be evaluated not only by how much CA1 activity changes, but also by whether the new population state remains interpretable at the output.

Comparing the two update rules clarifies why the form of rapid plasticity matters. The instructive-driven rule produced stronger one-shot reconstruction and a larger drop in tuning similarity at cue exchanges. The error-driven rule performed a bounded residual correction; after repeated track experience it produced greater spatial stability and much higher clean position accuracy. These differences should not be reduced to a single ranking. The instructive-driven rule is closer in spirit to an instructed write, whereas the error-driven rule adds a computational mechanism for correcting under- and overexpressed CA1 components. Because both rules used one shared parameter regime, their comparison reveals consequences of the update signal under matched conditions rather than the best performance each could achieve. A stronger model comparison would optimize each rule independently and vary memory load, cue overlap, and learning rate.

The degradation experiments distinguish robustness to partial input from mechanistic pattern completion. With the normal key map, recall remained informative after substantial input loss, and selective removal of MEC-like or LEC-like activity preferentially affected the corresponding readout. Yet the CA3-like population has no recurrent connections or attractor settling. Robustness arises from a distributed feedforward input-to-key map followed by a learned key-to-content association. Degraded-cue recall alone therefore does not identify a recurrent completion mechanism. A biological claim about CA3 pattern completion would require explicit recurrent dynamics and comparison with interventions that alter those dynamics [11, 17].

The dense common-key control also has a deliberately narrow interpretation. It forces all CA3-like units to receive the same EC average and thus destroys item-specific key structure while changing connection density and sparsity. Its failure shows that a non-specific common key cannot support these associations; it does not show that the selected sparse wiring is uniquely necessary or optimal. Likewise, the MEC/LEC dissociation is partly expected from the constructed input and readout partitions. Its value is as an internal validation that the shared CA1 basis can retain access to an intact component, not as evidence for strict anatomical modularity. Sparsity-matched controls would be needed to separate the effects of connection density, dimensionality, and key discriminability.

Several additional limitations define the scope of the work. The model omits dentate gyrus, recurrent CA3 dynamics, attractor competition, inhibition, consolidation, and explicit long-term drift of neurons or readouts. MEC and LEC are simplified input halves despite mixed selectivity in both regions. The CA1 basis is obtained from a shallow autoencoder rather than a biologically specified developmental process, and the plasticity equations are bounded abstractions rather than temporal or molecular models of BTSP. The compatibility manipulation is deliberately synthetic, and its matched-decoder rescue is an equivariance control rather than independent biological evidence. Finally, the track contains only two cue identities at one memory load; capacity, interference, and scaling across multiple cue-pair sequences remain unresolved.

These limitations suggest direct next tests. Model performance should depend more strongly on compatibility between an instructed CA1 state and its output basis than on the magnitude of CA1 change alone. Degraded-cue recall should track retrieval-key discriminability across sparsity-matched controls, and increasing the number of cue-pair sequences should eventually reveal an interference-dependent capacity limit. Experimentally, the broad prediction is that learning-related CA1 reconfiguration can remain behaviorally useful when recipient populations preserve or acquire an appropriate readout, whereas an equally large but readout-incompatible change should impair retrieval.

Within these bounds, the model offers a compact proposal linking a previously established representational basis, rapid instructed plasticity, retrieval-key-based selection, and downstream readability.

CA1 activity can change as a cognitive map is updated yet remain useful if retrieved content is expressed in coordinates that recipient circuits can interpret. Readout compatibility is therefore a useful axis for evaluating models of rapid hippocampal learning and for asking how a changing hippocampal code might support decodable experience over longer timescales.

4 Methods

4.1 Implementation, reproducibility, and statistical treatment

Simulations were implemented in Python using NumPy and PyTorch and run on CPU. All reported experiments used the same 20 root seeds (51001–51020). Conditions and plasticity rules within an experiment were paired by root seed. The two main experiment scripts write a resolved JSON configuration, compressed NumPy arrays, and a tidy CSV source-data table; plotting functions generate the individual panels from those saved arrays. The composed vector figures do not alter the underlying values.

During preliminary development, bounded hyperparameter configurations were explored using the covariance matrix adaptation evolution strategy (CMA-ES). This outer-loop engineering search used reconstruction- or recall-based objectives to locate workable regimes for population sparsity, activation gain, EC–CA3 convergence, and plasticity rate. The reported configurations were then fixed before the 20-seed comparisons. CMA-ES was not applied separately to individual seeds, experimental conditions, or plasticity rules; it is distinct from both Adam-based autoencoder pretraining and the local CA3–CA1 updates studied here. In generic form, the search proposed bounded parameter vectors θ and maximized a development-set score

$$J(\theta) = \frac{1}{N_{\text{dev}}} \sum_{n=1}^{N_{\text{dev}}} \cos\left(\hat{x}_{\text{ET}}^{(n)}(\theta), x_{\text{ET}}^{(n)}\right), \quad (1)$$

or its reconstruction-loss analogue. This procedure selected a computational operating point; it was not included as evidence for learning, remapping, or robustness, and optimization generations were not counted as experimental replicates.

The unit of computational replication was the root seed, not an individual memory, CA1 unit, track position, or corruption mask. Values were first averaged over observations nested within a seed and then summarized across seeds. For a seed-level statistic y_s , $s = 1, \dots, S$, curves and error bars show $\bar{y} \pm \text{SEM}(y)$, where

$$\bar{y} = \frac{1}{S} \sum_{s=1}^S y_s, \quad \text{SEM}(y) = \sqrt{\frac{1}{S(S-1)} \sum_{s=1}^S (y_s - \bar{y})^2}. \quad (2)$$

Confidence intervals (CIs) reported in the text are two-sided 95% Student- t intervals over the 20 seed-level values, $\bar{y} \pm t_{0.975, S-1} \text{SEM}(y)$. Contrasts use paired seed-level differences $d_s = y_s^{(A)} - y_s^{(B)}$ rather than differences between independently aggregated conditions. These intervals quantify sensitivity to sampled initializations and inputs, not uncertainty over a biological population. No null-hypothesis significance tests or multiple-comparison corrections were applied; all analyses are descriptive simulation results.

4.2 EC patterns and the pretrained CA1 basis

EC input, CA1, and CA3-like populations each contained 50 units. A one-hidden-layer autoencoder mapped EC input to a 50-unit CA1 representation and back to a 50-unit EC output. CA1 activity used a differentiable top- K sparse activation with $K_{\text{CA1}} = 5$ and inverse temperature $\beta = 25$; output activity used a sigmoid decoder with gain 25. Linear encoder and decoder biases were disabled. Specifically, if $z_{(1)} \geq \dots \geq z_{(N)}$ are the sorted preactivations, the activation threshold and sparse nonlinearity were

$$\vartheta_K(z) = \frac{z_{(K)} + z_{(K+1)}}{2}, \quad (3)$$

$$[\mathcal{S}_{K,\beta}(z)]_i = \sigma(\beta[z_i - \vartheta_K(z)]), \quad \sigma(u) = \frac{1}{1 + e^{-u}}. \quad (4)$$

Thus approximately K units lay above the adaptive half-activation threshold, while gradients remained defined near that threshold. For EC pattern $x_{\text{ET}} \in \{0, 1\}^{50}$, encoder E , and decoder D , pretraining computed

$$x_{\text{IS}} = \mathcal{S}_{5,25}(Ex_{\text{ET}}), \quad (5)$$

$$\tilde{x}_{\text{ET}} = \sigma(25Dx_{\text{IS}}), \quad (6)$$

$$\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_{n=1}^N \left\| \tilde{x}_{\text{ET}}^{(n)} - x_{\text{ET}}^{(n)} \right\|_2^2. \quad (7)$$

The weights were trained with Adam (learning rate 0.001, batch size 64) to minimize Eq. 7. Pretraining occurred before CA3–CA1 storage, after which the encoder and decoder were fixed. The encoder therefore supplied the instructed CA1 state (IS), and the decoder defined the output coordinates against which compatibility was tested.

For the readout-compatibility experiment, the autoencoder was trained for 1024 epochs on 2048 unique binary EC patterns with five active units and evaluated on 256 held-out patterns. Twenty-eight additional unique patterns were reserved for one-shot associative storage. For the cue-reconfiguration experiment, the autoencoder was trained for 256 epochs on the 40 training laps and evaluated on 10 separately generated laps. For the degradation experiment, it was trained for 256 epochs on 20 intact laps and evaluated on five separately generated laps. Each lap contained 50 positions.

Track inputs contained 25 MEC-like and 25 LEC-like units. For position $p \in \{0, \dots, L-1\}$ on a circular track of length $L = 50$, circular distance was $d_L(p, \mu) = \min(|p - \mu|, L - |p - \mu|)$. MEC unit i had a preferred position μ_i from an even tiling of the track and Gaussian activation probability

$$g_i(p) = \exp \left[-\frac{1}{2} \left(\frac{d_L(p, \mu_i)}{\sigma_{\text{MEC}}} \right)^2 \right], \quad \sigma_{\text{MEC}} = 5. \quad (8)$$

Binary MEC activity was sampled independently as $x_{\text{MEC},i}(p) \sim \text{Bernoulli}(g_i(p))$. Two fixed, non-overlapping 25-dimensional LEC cue templates q_1 and q_2 were centered at $c_1 = 10$ and $c_2 = 30$. Cue j was present with probability

$$g_j^{\text{cue}}(p) = \exp \left[-\frac{1}{2} \left(\frac{d_L(p, c_j)}{4} \right)^2 \right], \quad (9)$$

$$P_j(p) = \text{clip}_{[0,1]} \{ 2\sigma(40[g_j^{\text{cue}}(p) - 0.1]) - 1 \}. \quad (10)$$

If the Bernoulli cue-presence sample was positive, the LEC block was set to q_j ; otherwise it was zero. The complete entorhinal target was the concatenation $x_{\text{ET}}(p) = [x_{\text{MEC}}(p); x_{\text{LEC}}(p)]$. This stochastic construction generated different laps from the same underlying spatial and cue statistics.

4.3 CA3-like retrieval keys and CA3–CA1 plasticity

EC input was projected to 50 CA3-like units through a fixed sparse matrix followed by a differentiable top- K activation. The compatibility experiment used two EC inputs per CA3-like unit, $K_{\text{CA3}} = 5$, inverse-temperature parameter $\beta_{\text{CA3}} = 200$, and learning rate $\alpha = 0.08$. Track experiments used 29 EC inputs per CA3-like unit, $K_{\text{CA3}} = 3$, $\beta_{\text{CA3}} = 170.6$, and $\alpha = 0.092$. The normal track map was constructed to balance EC participation across CA3-like units. CA3–CA1 weights were initialized to zero. The CA1 recall activation used $K_{\text{CA1}} = 5$, with inverse temperature 25 in the compatibility experiment and 41.8 in the track experiments.

Writing and recall used the same fixed EC–CA3 projection A and frozen decoder D . At weight state W_t , the complete forward computation was

$$x_{\text{CA3}} = \mathcal{S}_{K_{\text{CA3}}, \beta_{\text{CA3}}}(Ax_{\text{ET}}), \quad (11)$$

$$x_{\text{CA1}} = \mathcal{S}_{5, \beta_{\text{CA1}}}(W_t x_{\text{CA3}}), \quad (12)$$

$$\hat{x}_{\text{ET}} = \sigma(25Dx_{\text{CA1}}). \quad (13)$$

Each selected input to a CA3-like unit had projection weight $1/50$; all other entries of A were zero. Consequently, the convergence parameter controlled which EC components contributed to a key, whereas the top- K nonlinearity controlled key population sparsity. Equation 13 was also used to compute the pre-update recall state required by the error-driven rule.

Let x_{ET} denote the EC input/target, x_{CA3} its retrieval key, x_{IS} the instructed CA1 state produced by the fixed encoder, and x_{CA1} the state recalled through the current CA3–CA1 weights W_t . The instructive-driven update was

$$W_{t+1} = (1 - \alpha x_{\text{IS}}) \odot W_t + \alpha x_{\text{IS}} x_{\text{CA3}}^{\text{T}}, \quad (14)$$

where the first term is broadcast across presynaptic dimensions. For the bounded error-driven rule,

$$e^+ = [x_{\text{IS}} - x_{\text{CA1}}]_+, \quad e^- = [x_{\text{CA1}} - x_{\text{IS}}]_+, \quad (15)$$

$$\delta^+ = \alpha(e^+ x_{\text{CA3}}^{\text{T}})(1 - W_t), \quad \delta^- = \alpha(e^- x_{\text{CA3}}^{\text{T}})W_t, \quad (16)$$

$$W_{t+1} = \text{clip}_{[0,1]}(W_t + \delta^+ - \delta^-). \quad (17)$$

Recalled CA1 activity was obtained by applying the same target sparsity to $W_t x_{\text{CA3}}$ and was passed through the fixed pretrained decoder. The instructive-driven rule depends on the instructed target and presynaptic key but not the current recall error. The error-driven rule separates the residual into positive and negative components, such that under- and overexpressed CA1 components drive bounded potentiation and depression, respectively. All products between equally shaped arrays were elementwise unless an outer product or matrix product is shown explicitly. Memories were presented sequentially, and W_{t+1} was carried to the next item or position without replay, batching, or gradient descent through the local plasticity dynamics.

4.4 Paired plasticity-rule design

Both rules were evaluated in every main experiment. Within a root seed they shared EC inputs, pretrained autoencoder weights, CA3 wiring seeds, storage order, and—in the degradation experiment—corruption masks. Each rule had its own CA3–CA1 weight matrix, which evolved according to the corresponding update. All memory hyperparameters were shared rather than optimized for each rule. Supplementary Fig. S1 collects four paired contrasts derived directly from the main result arrays; it is not a separate simulation.

4.5 Readout-compatibility experiment

For each seed, the 28 held-out EC patterns were assigned CA3-like retrieval keys and pretrained CA1 instructions, then written once in a shared random order. The aligned condition used the original IS and decoder. The fixed-permutation condition applied one seed-specific CA1 permutation to every IS while leaving the decoder unchanged. The matched-decoder condition used the same permuted instructions and correspondingly permuted decoder columns. The random-content condition associated each key with a shuffled memory instruction. The no-plasticity condition retained zero CA3–CA1 weights. Recall was evaluated using cosine similarity between reconstructed EC output and the corresponding EC target. Each seed’s plotted value is its mean across 28 memories.

More explicitly, let P be the permutation matrix and $h = x_{\text{IS}}$. The aligned condition stored h and read it with D ; the fixed-permutation condition stored Ph but retained D ; and the matched-decoder condition used Ph together with $D' = DP^T$. In the idealized exact-recall case,

$$D'Ph = DP^TPh = Dh, \quad (18)$$

so matched decoder relabeling restores the original output coordinates without restoring the original CA1 unit identities. This control isolates coordinate compatibility from a generic effect of permutation. Similarity between two nonzero vectors u and v was

$$\cos(u, v) = \frac{u^T v}{\|u\|_2 \|v\|_2}; \quad (19)$$

zero-norm cases were assigned similarity zero by the numerical implementation.

4.6 Cue-reconfiguration experiment

Models learned 40 complete laps. Cue identities exchanged positions every 10 laps, alternating between assignments $[0, 1]$ and $[1, 0]$. The matched no-swap schedule retained $[0, 1]$ for all 40 laps. Within a seed, the schedules shared the pretrained autoencoder, initial CA3–CA1 weights, MEC-like trajectory, lap count, and plasticity parameters; only cue identity after the first 10 laps differed by design.

After each training lap, plasticity was paused while both fixed cue contexts were probed; the probe matching the context scheduled for that lap was retained for the transition analysis. For each CA1 unit, activity was mean-centered across positions before calculating cosine similarity between consecutive spatial tuning curves. Similarities were then averaged across units. Cue-exchange transitions were compared with the same lap boundaries in the paired no-swap model.

Let $f_{\ell ci}(p)$ be the frozen-probe activity of CA1 unit i at position p , after training lap ℓ , under cue context c , and define the centered curve $\tilde{f}_{\ell ci}(p) = f_{\ell ci}(p) - L^{-1} \sum_{r=0}^{L-1} f_{\ell ci}(r)$. Consecutive-lap tuning similarity was

$$T_\ell = \frac{1}{N_{\text{CA1}}} \sum_{i=1}^{N_{\text{CA1}}} \cos(\tilde{f}_{\ell c_\ell i}, \tilde{f}_{\ell-1, c_{\ell-1}, i}). \quad (20)$$

The transition statistic compared T_ℓ at boundaries $\ell \in \{10, 20, 30\}$ with the same boundaries in the matched no-swap run.

After the final lap, plasticity remained frozen and the two cue contexts were probed. For each CA1 unit, spatial stability was the cosine similarity between its two mean-centered context tuning curves. Cue modulation was the mean absolute activity difference between contexts. These metrics quantify context sensitivity but do not themselves classify rate or global remapping. Heat maps show the error-driven seed whose mean stability and modulation were closest to the across-seed medians; units were ordered by preferred position across the two contexts. For final-lap contexts a and b , the plotted unit-level quantities were

$$S_i = \cos(\tilde{f}_{ai}, \tilde{f}_{bi}), \quad (21)$$

$$M_i = \frac{1}{L} \sum_{p=0}^{L-1} |f_{ai}(p) - f_{bi}(p)|. \quad (22)$$

Here S_i measures preservation of spatial profile after removing mean firing, whereas M_i retains absolute context-dependent rate differences; their joint distribution therefore separates two complementary aspects of mixed tuning.

4.7 Selective degradation experiments

Models stored 20 intact laps with cue assignment $[0, 1]$. During frozen recall, a fixed random subset of eligible EC units was zeroed throughout a complete probe lap. Masks were sampled independently between probes. Dropped fractions were 0, 0.25, 0.50, 0.75, and 0.90, with 12 masks per fraction and root seed. LEC-like degradation operated on the final 25 input units; MEC-like degradation operated on the first 25.

For dropped fraction q and mask replicate r , a binary mask $m^{(q,r)} \in \{0, 1\}^{50}$ set $\text{round}(25q)$ randomly selected coordinates in the targeted input half to zero and left all other coordinates at one. The same mask was held fixed for all positions of that probe lap:

$$x_{\text{ET}}^{(q,r)}(p) = m^{(q,r)} \odot x_{\text{ET}}(p). \quad (23)$$

Plasticity was disabled during these probes, so degradation altered retrieval evidence but did not rewrite the stored association.

Cue identity was decoded at the two cue positions by cosine comparison between recalled LEC output and the two fixed cue templates. Spatial accuracy was the fraction of recalled MEC outputs whose nearest clean MEC target had the correct track-position index; chance was therefore $1/50$. MEC output cosine was averaged across all positions. For every curve, the 12 mask-level values were first averaged within a seed before means and uncertainty were calculated across seeds.

Writing $\hat{x}_{\text{LEC}}(c_j)$ for recalled LEC output at cue center c_j and $x_{\text{MEC}}^*(p)$ for the clean MEC target at position p , the three reported metrics were

$$A_{\text{cue}} = \frac{1}{2} \sum_{j=1}^2 \mathbf{1} \left[j = \arg \max_{k \in \{1,2\}} \cos(\hat{x}_{\text{LEC}}(c_j), q_k) \right], \quad (24)$$

$$A_{\text{pos}} = \frac{1}{L} \sum_{p=0}^{L-1} \mathbf{1} \left[p = \arg \max_{u \in \{0, \dots, L-1\}} \cos(\hat{x}_{\text{MEC}}(p), x_{\text{MEC}}^*(u)) \right], \quad (25)$$

$$C_{\text{MEC}} = \frac{1}{L} \sum_{p=0}^{L-1} \cos(\hat{x}_{\text{MEC}}(p), x_{\text{MEC}}^*(p)). \quad (26)$$

Cue accuracy has two-class chance level $1/2$, whereas exact-position accuracy has chance level $1/L = 1/50$. MEC cosine is graded and has no corresponding uniform-classification chance line.

The dense common-key control replaced the EC–CA3 map with a matrix whose every entry was $1/50$. All CA3-like units consequently received the same average EC input and generated a common, non-item-specific key. Because this control also changes connection density and sparsity relative to the normal map, it tests the need for key discriminability rather than the isolated effect of sparse connectivity. In matrix notation the control used $A_{ij} = 1/50$ for all i, j ; hence every CA3 preactivation was identical for a given input. Equation 4 therefore returned the same 0.5-valued vector for every input, rather than an item-specific key.

Data and code availability

The simulations underlying Figs. 2 and 3 are generated by the scripts in `src/experiments`; their resolved configurations, compressed numerical arrays, and tidy source-data tables are stored in `results/preprint`. A permanent public archive and accession identifier will be added before public posting.

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Supplementary material

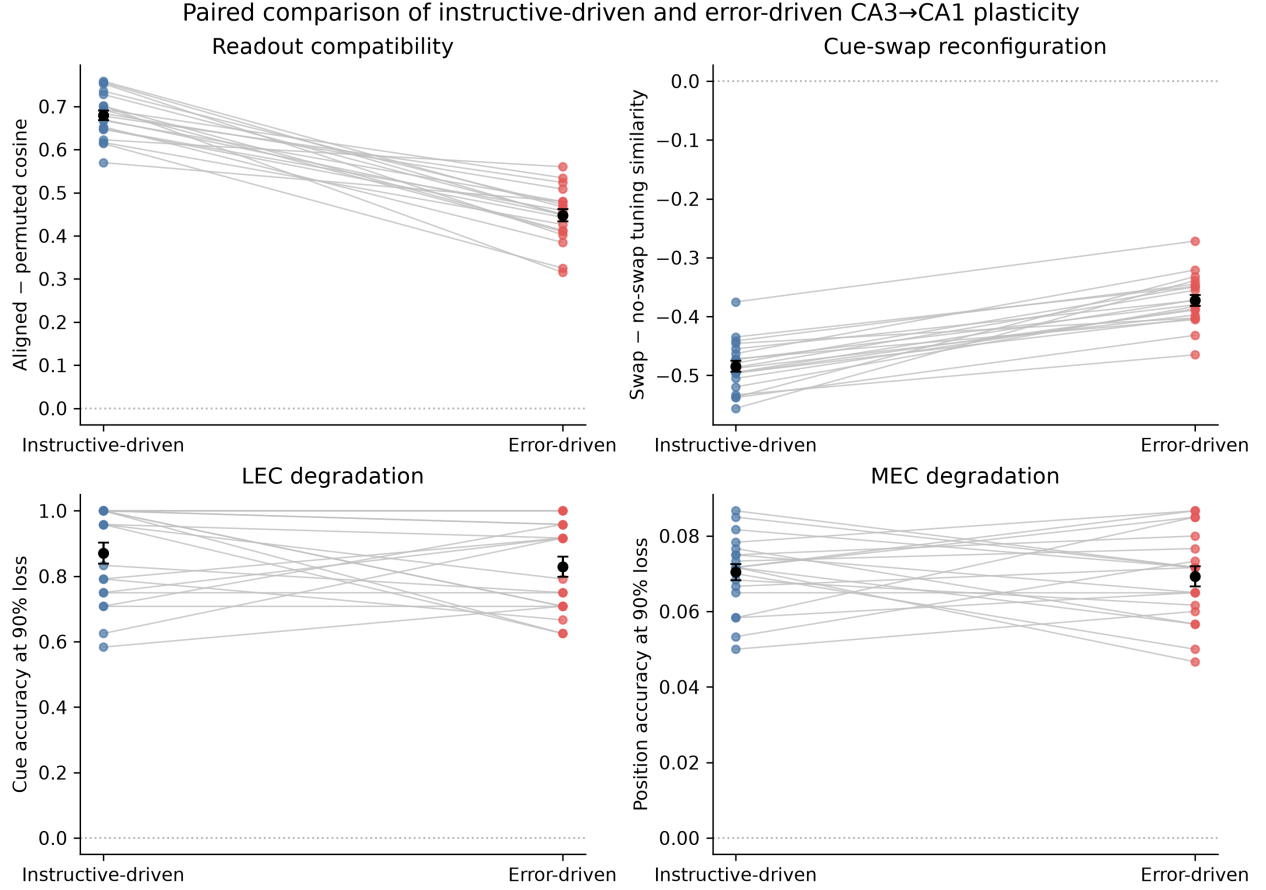


Figure S1: Seed-paired comparison of instructive-driven and error-driven plasticity. Each grey line joins instructive-driven and error-driven values obtained from the same root seed; colored points are individual seed-level values and black points with error bars show mean $\pm$ SEM across 20 seeds. **(Upper left)** Readout-compatibility effect, defined as aligned minus fixed-permutation output-target cosine in Fig. 2A. **(Upper right)** Cue-exchange effect, defined as tuning similarity at exchange boundaries minus similarity at the same boundaries in the no-swap control in Fig. 2B. **(Lower left)** Cue identity accuracy after 90% LEC-like degradation in Fig. 3A. **(Lower right)** Nearest-position accuracy after 90% MEC-like degradation in Fig. 3C. Both rules shared inputs, pretrained autoencoders, CA3 wiring seeds, storage order, corruption masks, and hyperparameters; only the CA3-CA1 update differed. Parameters were not optimized separately for each rule.